MIL-R-25565E
13 August 1974
SUPERSEDING
MIL-R-25565D
9 June 1970

MILITARY SPECIFICATION

RELEASES, PARACHUTE RIPCORD, AUTOMATIC, TYPES F-1B AND FF-1

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers two types of automatic, parachute-ripcord releases, designated Types F-1B and FF-1.

1.2 Classification. The automatic, parachute-ripcord release shall be one of the following classes, as specified (see 6.2):

Class 1 - see Figure 1
Class 2 - see Figure 1
Class 3 - see Figure 1
Class 4 - see Figure 2
Class 5 - see Figure 3

2. APPLICABLE DOCUMENTS

* 2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATIONS

Federal

QQ-N-290	Plating, Nickel (Electrodeposited)
PPP-B-601	Boxes, Wood, Cleated-Plywood
PPP-B-636	Boxes, Shipping, Fiberboard
PPP-B-676	Boxes, Setup

Military

MIL-P-116	Preservation, Methods Of
MIL-S-5059	Steel, Corrosion Resistant (18-8), Plate, Sheet And Strip
MIL-E-5272	Environmental Testing, Aeronautical And Associated Equipment, General Specification For
MIL-W-5424	Wire Rope: Steel (Corrosion-Resisting), Flexible, Preformed (For Aeronautical Use)
MIL-A-8625	Anodic Coatings, For Aluminum And Aluminum Alloys

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T06:42Z
Check the source to verify that this is the current version before use.

STANDARDS

Federal

FED-STD-595	Colors

Military

MIL-STD-100	Engineering Drawing Practices
MIL-STD-105	Sampling Procedures And Tables For Inspection By Attributes
MIL-STD-129	Marking For Shipment And Storage
MIL-STD-130	Identification Marking Of U.S. Military Property
MIL-STD-143	Standards And Specifications, Order Of Precedence For The Selection Of
MIL-STD-202	Test Methods For Electronic And Electrical Component Parts
MIL-STD-831	Test Reports, Preparation Of
MIL-STD-889	Dissimilar Metals
MIL-STD-1186	Cushioning, Anchoring, Bracing, Blocking, And Waterproofing; With Appropriate Test Methods
MS20664	Ball End, Wire Rope, Swaging, Single Shank
MS27039	Screw, Machine - Pan Head, Structural, Cross Recessed
MS70104	Housing, Parachute Ripcord

Air Force-Navy Aeronautical

AN936	Washer, Tooth Lock
AN3054	Nut, Coupling, Electrical Conduit

DRAWINGS

Air Force

55C6448	Lanyard - Automatic Parachute Ripcord Release, Installation Of
57C6109	Lanyard - Automatic Parachute Ripcord Release Installation Of
59B6223	Snap Hook - Parachute Lanyard, Assembly Of
62A4261	Anchor - Parachute Ripcord Release Automatic
62C4266	Lanyard - Automatic Parachute Ripcord Release, Assembly Of
66C1521	Guide-Parachute Ripcord Grip

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T06:42Z
Check the source to verify that this is the current version before use.

U. S. Gauge Co.

AF-1104	Holder-Arming Pin
AF-1105	Washer-Arming Pin
AF-1188	Screw-Power Cable Guide
AF-1211	Washer-Power Cable Guide
AF-1218	Pin Arming
AF-1219	Arming Knob Assembly
AF-1221	Ferrule-Arming Knob
AF-1226	Arming Cable Housing Ass'y
AF-1231	Army Pin Housing Ass'y
AH-1834-AD	Ring Retaining

Anaconda Metal Hose Division

79790	Housing-Ripcord Power Cable

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Qualification. Releases furnished under this specification shall be products which are qualified for listing on the applicable qualified products list at the time set for opening of bids (see 4.3 and 6.3).

3.2 Components. The release shall consist of the following major components:

Item No.	Quantity	Item
1	1	Power source
2	1	Timer mechanism
3	1	Aneroid mechanism
4	1	Ripcord power cable assembly
5	1	Arming assembly
6	1	Cover

3.3 Selection of specifications and standards. Specifications and standards for necessary commodities and services not specified herein shall be selected in accordance with MIL-STD-143.

3.4 Materials.

3.4.1 Fungus-proof materials. Materials that are nutrients for fungi shall not be used where it is practical to avoid them. Where used and not hermetically sealed, they shall be treated with a fungicidal agent acceptable to the procuring activity. However if they will be used in a hermetically sealed enclosure, fungicidal treatment will not be necessary.

3.4.2 Metals. Metals shall be of the corrosion-resistant type or suitably treated to resist corrosion due to fuels, salt spray, or atmospheric conditions likely to be met in storage or normal service.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T06:42Z
Check the source to verify that this is the current version before use.

* 3.4.2.1 Dissimilar metals. Unless suitably protected against electrolytic corrosion, dissimilar metals shall not be used in intimate contact with each other. Dissimilar metals are defined in MIL-STD-889.

3.5 Design and construction. The release shall be designed to pull the ripcord of a parachute after a time delay. The time delay shall be presettable as described in 3.8. The design shall include a pressure sensitive device which prevents the release from operating above a preset altitude (see 3.9). The release shall be constructed as shown in Figures 1, 2 and 3. The mechanism shall be a complete unit, self-contained and separate from the case assembly. The mechanism shall be compatible with the various case configurations. The case shall be constructed to protect the mechanism. With the cover removed, the case shall exclude dust and particles capable of jamming the mechanism.

3.5.1 Adjustment and repairs. The release shall be so constructed that no parts will work loose in service. It shall be built to withstand the strains, jars, vibrations, and other conditions incident to shipping, storage, installation, and service.

3.5.2 Vibration proofing. All nuts, bolts, cover plates, et cetera. shall be safetied with wire, sealed with a satisfactory sealing compound, or permanently retained in some other manner to prevent loosening from vibration.

3.5.3 Cocking. The release shall be so designed that, after use, it can be cocked for further use without the aid of special tools, fittings, or attachments, except a winding key. The cocking procedure shall be as simple as possible and shall require the services of only one man. One winding key shall be provided with each tenth device. The winding key shall be so designed that it can be used to engage or disengage the center holddown screw of the reinforced cover assembly.

3.6 Performance. The release shall be capable of operating satisfactorily as follows:

a. At temperatures ranging from -54° to +71° Centigrade (C).

b. After exposure to relative humidity up to 100 percent, including conditions where condensation takes place in the form of both water and frost.

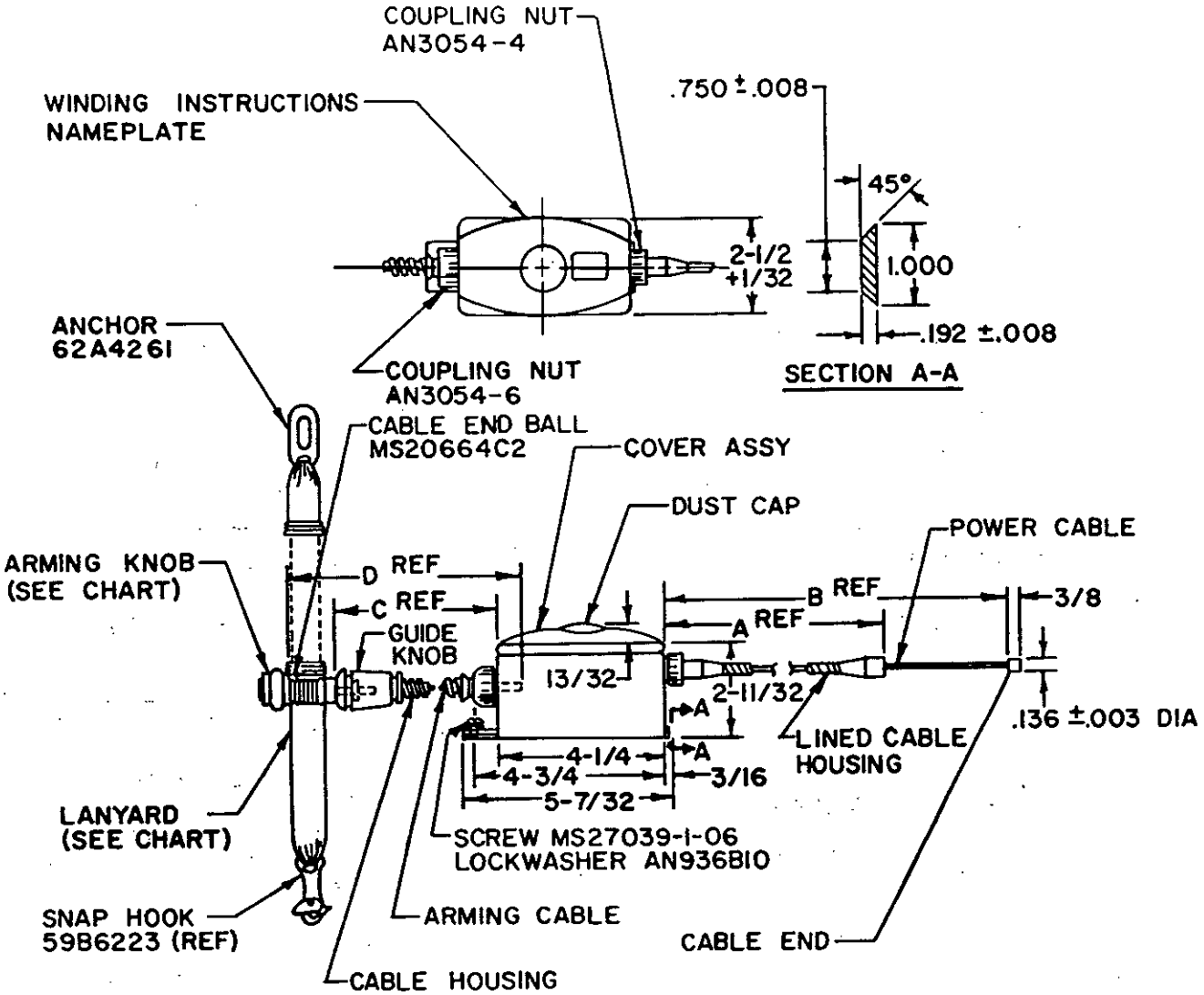
c. For type F-1B, at pressures ranging from 30 inches mercury down to 13.75 inches mercury (approximately an altitude of 20,000 feet), and after exposure to altitude conditions of 50,000 feet (3.4 inches mercury). For type FF-1, at pressures ranging from 30 inches mercury down to 16.88 inches mercury (approximately an altitude of 15,000 feet) and after exposure to altitude conditions of 50,000 feet (3.4 inches mercury).

d. After exposure to salt-sea atmosphere.

e. After exposure to sand and dust particles as may be encountered in desert areas.

f. Under acceleration and shock forces incident to service use.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T06:42Z
Check the source to verify that this is the current version before use.



DIMENSIONS IN INCHES
UNLESS OTHERWISE SPECIFIED
TOLERANCES: FRACTIONS ±1/64
ANGLES ±1°

RELEASE	A ±1/8	B ±1/8	C ±1/8	D ±3/16	LANYARD	KNOB
CLASS 1	10-1/16	16-3/4	14	16	57C6109	FIG 4 (REF)
CLASS 2	10-1/16	16-3/4	14	16	55C6448	FIG 4 (REF)
CLASS 3	6-9/16	12	22	24	N/A	FIG 4 (REF)

FIGURE 1. RELEASE, CLASSES 1, 2, AND 3

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T06:42Z
Check the source to verify that this is the current version before use.

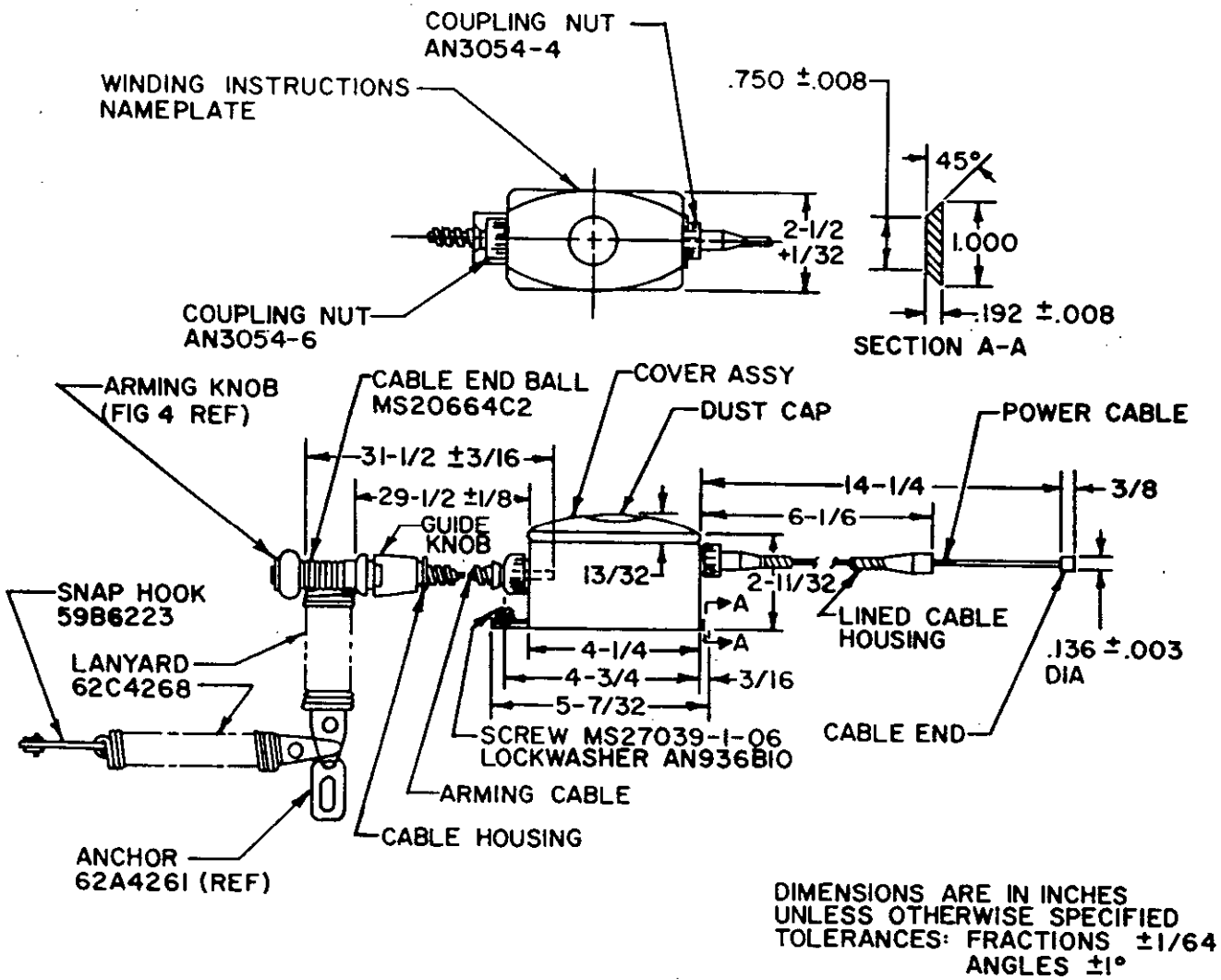


FIGURE 2. RELEASE, CLASS 4

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T06:42Z
Check the source to verify that this is the current version before use.

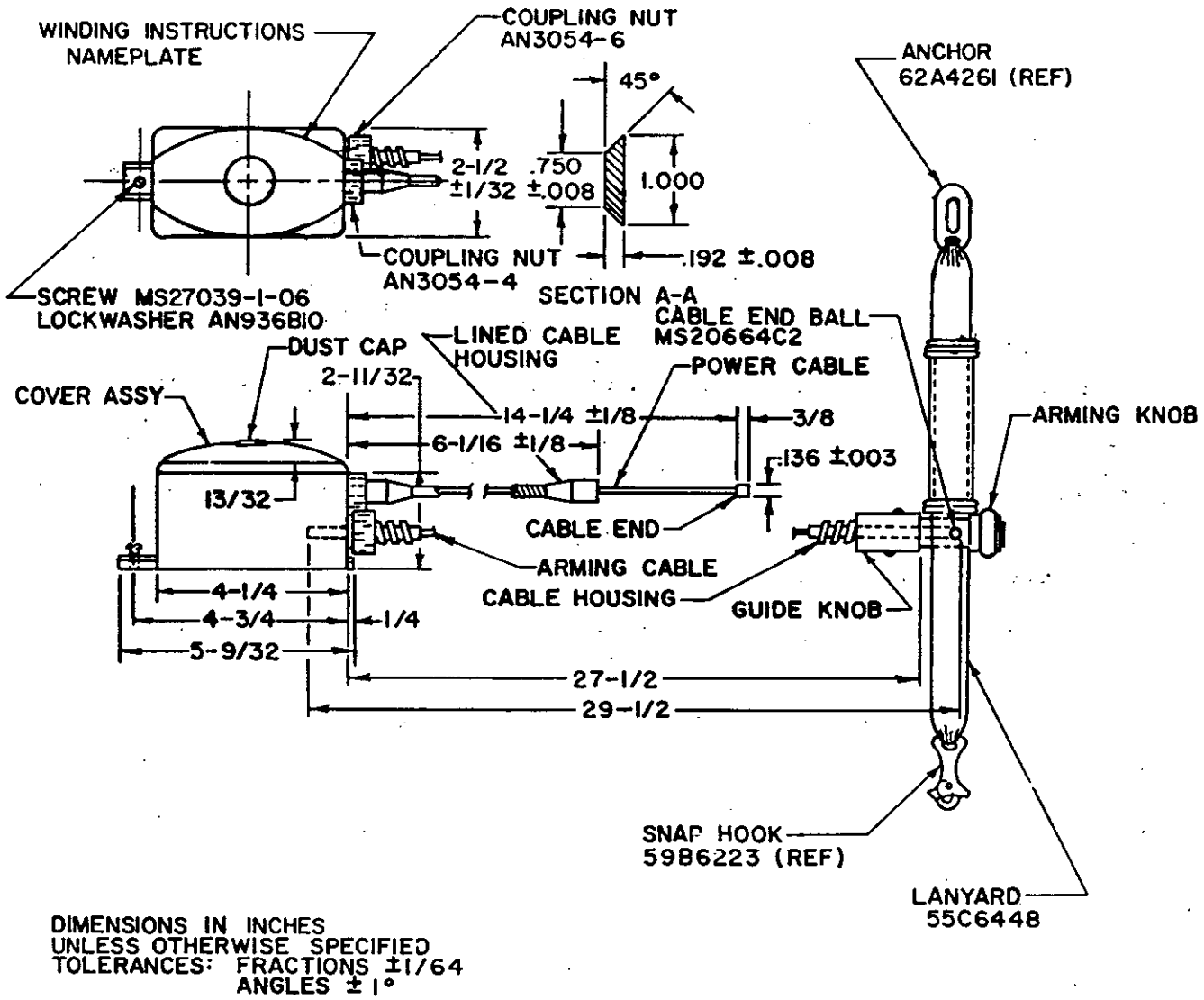


FIGURE 3. RELEASE, CLASS 5

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T06:42Z
Check the source to verify that this is the current version before use.

g. Under vibration conditions incident to service use.

h. After an overpressure of 50 inches mercury.

i. At human free-fall velocities below 25,000 foot pressure altitude.

3.6.1 Pressure differential. The pressure within the case shall at all times be within ±100 feet of the pressure surrounding the case.

* 3.6.2 Pressure sensing device accuracy. The pressure sensing device mechanism shall maintain the accuracy specified in 3.9 after 1,000 cycles of varying the pressure from sea level to 35,000 feet.

3.6.3 Knob strength. With the knob used as the point of support, it shall be capable of withstanding a 100-pound weight suspended from the arming pin.

* 3.6.4 Life. The release shall be capable of performing satisfactorily after it has been operated 70 times, without visible damage to the power cable.

3.7 Power source. The release shall be wholly mechanical and shall be powered by a coil spring with a minimum pull on the ripcord of 45 pounds after 2 inches of cable movement with the timer set for 2.5 seconds, and 36 pounds after 2 inches of cable movement with the timer set for 12-1/2 seconds. The total movement of the cable shall be a minimum of 4 inches. The cable must be confined on the power drum in order that no damage to the cable or mechanism will result if fired without a load. The release shall be so designed that the ripcord action cable cannot be extended and the timing mechanism cannot be set unless the power spring is fully wound. A positive means shall be provided to prevent the disengagement of the cable drum from the power unit under conditions of acceleration

3.8 Timer mechanism. The timer control shall be so designed that it can be set throughout the range of 1 to 12-1/2 seconds in 1/2 second increments. It shall not be set at 1/2 second. The time scale and setting mechanism shall be so designed that the time setting may be quickly and accurately made, and it shall be impossible to set the mechanism at any interval other than the 1/2-second intervals between 1 and 12-1/2 seconds. The accuracy shall be within ±0.2 second up to the 2.5-second setting and ±10 percent of each setting throughout the remainder of the range. This accuracy shall apply for all temperatures from -54° to +71°C. The escapement shall be hardened to reduce wear and shall be positively started when, and only when, the arming pin is withdrawn. The release shall be so designed that the timer mechanism cannot be set or the power cable extended unless the power spring is fully wound.

* 3.9 Pressure sensing device mechanism.

a. For Type F-1B the function of the pressure sensing device mechanism is to block the timer escapement and prevent operation of the release at all altitudes approximately 1,500 feet above that indicated by the altitude dial setting. The altitude-setting dial shall be marked in increments of 1,000 feet between 5,000 and 20,000 feet, with each 5,000-foot point being identified with a numeral as well as a graduation. It shall be feasible to change the setting with a gloved hand. As a safety measure, it shall be possible to insert the arming pin regardless of the setting of the pressure sensing device dial. The pressure sensing device shall be calibrated to release the timer escapement at the pressure altitudes listed in Table I for the various altitude setting

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T06:42Z
Check the source to verify that this is the current version before use.

points. The purpose of this is to cause the ripcord to be pulled at approximately the preset altitude, taking into consideration the fact that the man is still free-falling and thus losing altitude during the operation of the timer mechanism prior to tripping the release. The altitude-setting scale shall be as long as possible, but shall be no less than 1.46 inches, or an arc of 135 degrees included angle with a radius of 5/8 inch.

TABLE I. Pressure Altitudes For Type F-1B

Pressure sensing device dial setting (feet)	Pressure altitude above which timer is blocked	Tolerances in feet		
		-54°C	Room	+71°C
5,000	6,500	±1,000	±500	±1,000
10,000	11,500	±1,000	±750	±1,000
14,000	15,500	±1,000	±800	±1,000
20,000	21,500	±1,000	±900	±1,000

b. For Type FF-1 the altitude setting dial shall be marked in increments of five hundred (500) feet between 3,000 and 15,000 feet, with the 3,000, 5,000, 10,000 and 15,000 foot points being identified with a numeral as well as a graduation. Intermediate graduations to be 4, 6, 7, 8, 9, 11, 12, 13 and 14 thousand feet and minor graduations at five hundred (500) foot increments. A positive block shall be provided at the minimum aneroid altitude setting (3,000 feet) to insure that the altitude dial indicator cannot be inadvertently set below the intended minimum setting. It shall be feasible to change the setting with a gloved hand. As a safety measure, it shall be possible to insert the arming pin regardless of the setting of the altitude dial indicator. With the arming pin inserted, the pressure sensing device cam contact arm shall be sufficient distance from the pressure sensing device cam to prevent actual contact during the application of vibration, acceleration and shock force. The pressure sensing device shall be calibrated to release the timer escapement at the pressure altitudes listed in the Table II for the various altitude setting points, and maintain linear accuracy at the intermediate and minor graduations to the maximum feasible. It shall be possible to arm the release 700 feet above pressure altitude at which the timer is blocked as determined in 4.5.4.2 without the release activating. The span of the dial shall be 180 degrees plus or minus sufficient tolerance for accurate calibration to the pressure sensing device graduations cited above. The pressure sensing device guide pin shall be installed in a manner that will insure stable aneroid position and positive retention of the guide pin.

TABLE II. Pressure Altitudes for Type FF-1

Pressure sensing device Dial Setting (ft)	Pressure Altitude Above Which Timer Is Blocked	Tolerance in Feet		
		-54°C	Room	+71°C
3,000	4,500	±500	±500	±500
5,000	6,500	±500	±500	±500
10,000	11,500	±1,000	±750	±1,000
15,000	16,500	±1,000	±900	±1,000

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T06:42Z
Check the source to verify that this is the current version before use.

* 3.9.1 Pressure sensing device leak indicator. A simple, pressure sensing device leak indicator shall be incorporated to warn inspecting personnel of a badly leaking sensing device. The indicator shall be located so that it will be readily visible when the top cover is removed. The mechanism shall consist of an indicator actuated by the pressure sensing device to constantly show the altitude to which it is exposed. It shall provide graduations in 1,000-foot intervals between 3,000 and 25,000 feet and shall show a band indicating sea level at all altitudes, under 3,000 feet. This altitude indicator shall be calibrated within ±1,000 feet when subjected to an altitude of 5,000 feet and ±3,000 feet at any other altitude. If the indication and the leak detector differ by more than 3,000 feet from the known altitude of the observer, a leak and the magnitude of the leak shall be indicated.

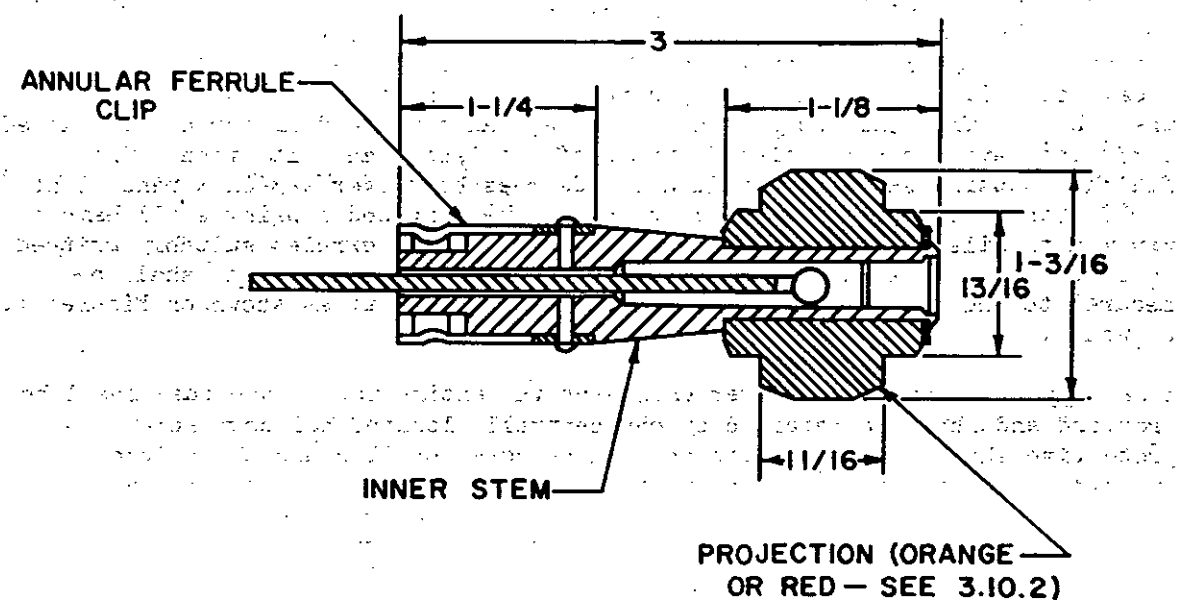
3.10 Arming assembly.

* 3.10.1 Arming cable. Arming of the release shall be accomplished through the withdrawal of a stainless-steel pin, 0.132 ±0.005 inch in diameter and 1-1/4 ±1/64 inch in length. The arming pin shall be swaged to a stainless-steel cable, 1/16 inch in diameter conforming to MIL-W-5424, and of the length specified by the procuring activity from Figure 1 (classes 1, 2, and 3), Figure 2 (class 4), or Figure 3 (class 5). A chamfer of 45 degrees shall be put on the front end of the arming pin and a 15 degree chamfer blending to an approximate 0.025-inch radius on the cable end of the pin. When inserted in the release, the arming pin shall be safetied or retained in such a manner that a force of between 10 and 20 pounds in tension shall be required to remove the arming pin. Opposite the end which bears the arming pin, the arming cable shall terminate in a steel ball conforming to MS20664.

* 3.10.2 Arming knob. The arming knob shall be constructed to the approximate outline shown on Figure 4. Material for the inner stem shall be aluminum. High impact plastic material will be suitable for the serrated annular concentric projection of the knob. Finish and color of the knob projection shall be international orange color 12197 conforming to FED-STD-595 or insignia red color 509 conforming to FED-STD-595. The arming knob, cable, coil spring, fabric strap, and hook used as an assembly shall be capable of withstanding a minimum static load of 350 pounds. The annular ferrule clip of the arming knob shall snap over the ferrule of the arming cable housing with sufficient retention to insure that a force of 15 to 20 pounds is required to remove the arming knob from the ferrule. Repeated use shall not reduce this force. The arming knob case shall be assembled to the arming cable and the length and dimensions of both arranged so that the arming knob case will be removed from the ferrule of the housing a distance of at least 1 inch before any tension is exerted on the arming cable. The arming cable lengths shall be as specified by the procuring activity from lengths listed on Figure 1, 2, or 3 (see 6.2).

3.10.3 Cable housing and ferrules. The cable housing shall be furnished by the contractor and shall be in accordance with MS70104 except that the ferrule and grommet at one end shall be such that it can be firmly retained by an AN3054-6 conduit nut as shown on Figures 1, 2, and 3. The length shall be as specified by the procuring activity from Figure 1 (classes 1, 2 and 3), Figure 2 (class 4), or Figure 3 (class 5). The finished housing shall withstand a proof load in tension of 150 pounds with the end ferrules suitably gripped in the fixture in order that the load may be applied uniformly. This tensile-strength requirement applies to the end ferrules only.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T06:42Z
Check the source to verify that this is the current version before use.



DIMENSIONS IN INCHES
UNLESS OTHERWISE SPECIFIED

FIGURE 4. APPROXIMATE OUTLINE—ARMING KNOB WITHOUT RETRACTABLE LANYARD

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T06:42Z
Check the source to verify that this is the current version before use.

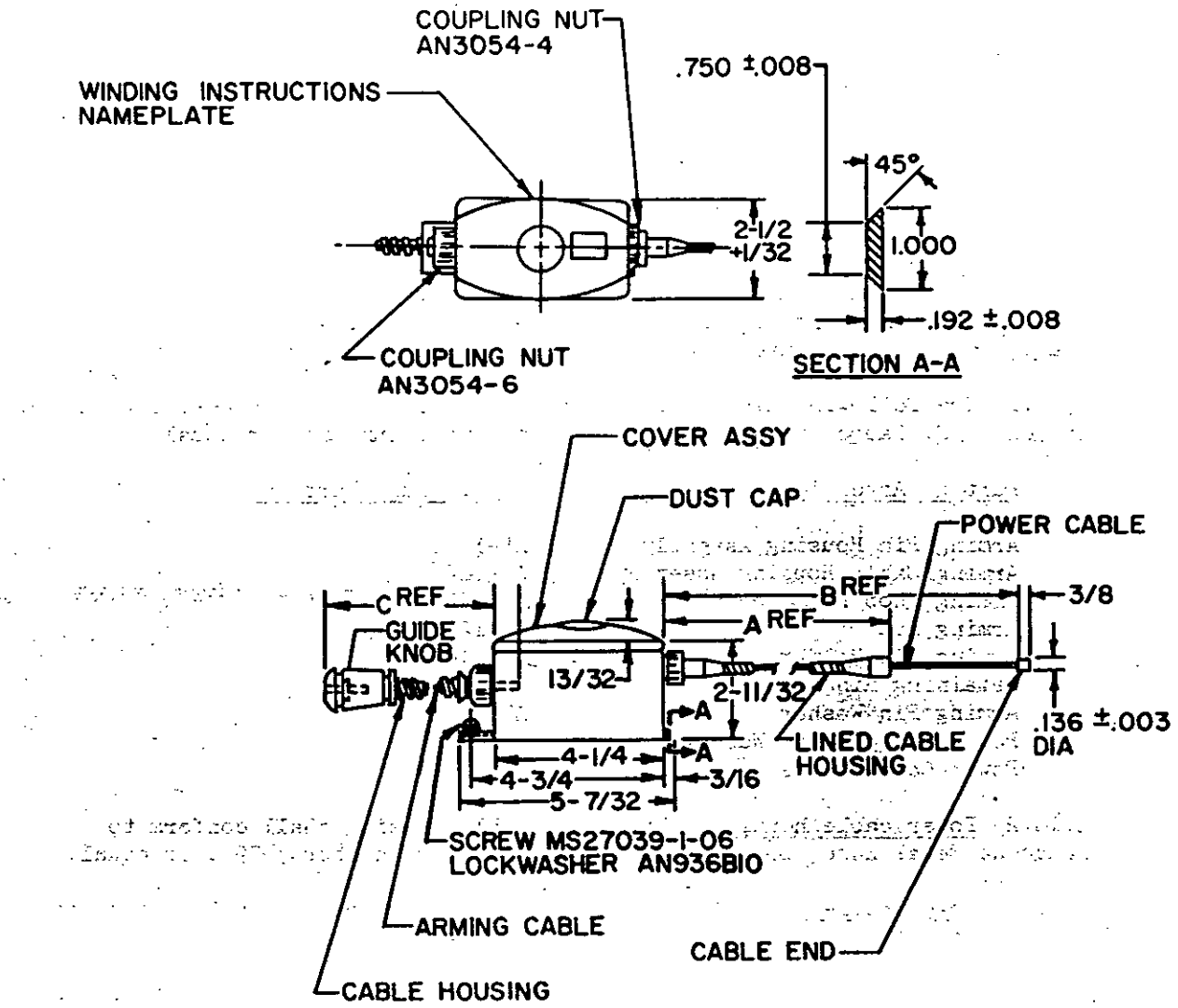
* 3.10.4 Knob guide. The knob guide shall conform to Drawing 66C1521, except that for the FF-1 release A knob guide conforming to the configuration illustrated in Figure 5 shall be provided. The guide shall be supplied with two fillister head screws of a length sufficient to firmly retain the cable housing as specified in 3.10.3. The screws shall be secured with lockwashers concealed in the counterbored holes in the knob guide.

* 3.11 Ripcord power cable assembly. The ripcord power cable shall consist of a 3/64-inch diameter, 7 by 7 construction, stainless steel cable, constructed and tested in accordance with MIL-W-5424. The outer end of the power cable shall have a swaged termination of the length and the diameter shown on Figures 1, 2, and 3. This swaging shall meet the strength requirements of MS20664. The length of the power cable and power cable housing shall be as shown on Figures 1, 2, and 3; the class shall be designated by the procuring activity (see 6.2). The contractor shall furnish the 1/4-inch inside diameter, flexible, stainless steel conforming to MIL-S-5509 (American Iron and Steel Institute's type 302), tubing, commercially designated as unpacked, interlocked construction. This tubing shall be fabricated by the spiral winding of formed strip stock. It shall be terminated in stainless steel fittings as shown on Figures 1, 2 and 3. The tubing shall be internally lined with Teflon, or equal, of a 0.020 inch minimum thickness. The lining shall adhere snugly to the tubing. The finished housing shall have an outside diameter not to exceed 0.375 inch and a clear inside diameter of not less than 0.180 inch. The finished housing shall withstand a 180 degrees bend over a 3-inch mandrel at -54°C without cracking or visible damage. The finished housing shall have a minimum tensile strength of 150 pounds with the end ferrules suitably gripped in fixtures so that the load is applied uniformly. This housing shall be secured to the case by an AN3054-4 conduit coupling nut as shown on Figures 1, 2, and 3.

3.12 Cover. A one-piece cover enclosing the entire top of the case shall be provided and shall be retained by one centrally located holddown screw. A protective shield suitably attached to the cover shall house the holddown screw and form a cover assembly. The protective shield shall be 13/32 ±1/16 inch in thickness. The cover assembly shall withstand a crush load of 800 pounds minimum applied in a perpendicular direction to any portion of the top of the cover without affecting the operating performance of the release. The holddown screw shall be free to rotate in the cover assembly but shall be so designed that in normal usage it will remain with the cover at all times. The holddown screw shall also be so designed that upon disengagement from the center stud, it shall be mechanically retained in a fully retracted position until re-engaged with the center stud. The holddown screw and cover assembly shall be designed to permit the use of a 10-cent piece to engage or disengage the holddown screw. An access hole to the holddown screw shall be employed in the design and shall be sealed dusttight with a readily removable cap. A gasket located in a retaining groove shall seal the joints at the holddown screw and case. The cover assembly shall be so designed that it cannot be assembled to the case when the timing knob is not fully seated. The cover assembly shall be of such rigid construction as to prevent forcing or warping and thereby giving a false indication of proper assembly.

* 3.12.1 The pressure sensing device mechanism shall be protected by a separate protective shield so that dust will be retarded and particles capable of jamming the mechanism will be isolated from the gear train and escapement area even though the case cover is removed.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T06:42Z
Check the source to verify that this is the current version before use.



DIMENSIONS IN INCHES
UNLESS OTHERWISE SPECIFIED
TOLERANCES: FRACTIONS ±1/64
ANGLES ±1°

FIGURE 5. FF-1 RELEASE

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T06:42Z
Check the source to verify that this is the current version before use.

3.13 Anchor. Unless otherwise specified, one anchor conforming to Drawing 62A4261 shall be provided with each release as shown on Figures 1, 2, and 3. The anchor shall be attached at the retractable lanyard as indicated on Figures 1, 2, and 3.

3.14 Snap hook lanyard. Unless otherwise specified, one snap hook conforming to Drawing 59B6223 shall be provided with each release as shown on Figures 1, 2 and 3.

3.15 Lanyard. Unless otherwise specified, one lanyard shall be provided with each release as shown on Figures 1, 2, and 3.

3.16 Part numbering of interchangeable parts. All parts having the same manufacturer's part number shall be functionally and dimensionally interchangeable. The item identification and part number requirements of MIL-STD-100 shall govern manufacturer's part numbers and changes thereto.

3.16.1 The following assemblies and parts shall be interchangeable with the listed U. S. Gauge Division of AMETEK, Inc. part numbers, or equal:

Part of Assembly	U. S. Gauge Part No.
Arming Pin Housing Assembly	AF-1231
Arming Cable Housing Assembly	AF-1226
Arming Knob Ferrule	AF-1221
Arming Pin	AF-1218
Arming Pin Holder	AF-1104
Retaining Ring	AH-1834-AD
Arming Pin Washer	AF-1105
Power Cable Guide Washer	AF-1211
Power Cable Guide Screw	AF-1188

3.16.2 Power cable housing. The power cable housing shall conform to Anaconda Metal Hose, Division of America Co. part number 79790, or equal.

3.16.3 Power cable connector. The power cable connector shall conform to Jay El Products, Inc. part number 3045-100, or equal.

3.17 Dimensions. The dimensions of the F-1B release shall be in accordance with Figures 1, 2, or 3, depending on class. The dimensions of the FF-1 shall be in accordance with Figure 1, class 3.

3.18 Weight. The weight of the release, including the ripcord power cable assembly and the arming assembly, shall not exceed 2.90 pounds.

3.19 Operation markings.

3.19.1 Setting identification. The words ALTITUDE SETTING FEET X 1,000 and TIMER SETTING IN SECONDS shall be permanently and legibly affixed in letters as large as possible on or near the altitude and time-setting knobs, respectively.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T06:42Z
Check the source to verify that this is the current version before use.

3.19.2 Winding instructions. An instruction plate, permanently and legibly marked with the following information, shall be securely attached to one long side of the case and may be combined with the nameplate, if desired:

WINDING INSTRUCTIONS

INSERT ARMING PIN - LIFT TIMER KNOB AND SET AT 1 SECOND
WIND FULLY WITH FORCE (INTERNAL STOP PREVENTS OVERWINDING)
LIFT TIMER KNOB BEFORE AND WHILE EXTENDING POWER CABLE
SET TIME

3.19.3 A decal shall be affixed to the cover of all classes of releases in the location shown on Figures 1, 2 and 3. The following instruction shall be printed on the decal:

DO NOT
FORCE COVER

TIMER KNOB
MUST BE SEATED

3.20 Finishes and protective coatings. All moving parts which can be so treated shall be electropolished or passivated if made from stainless steel, electroplated in accordance with QQ-N-290 if made from brass or bronze, or anodized in accordance with MIL-A-8625 if made from aluminum alloys.

3.20.1 The outer case and cover of the FF-1 release shall be yellow, Shade 13655 of FED-STD-595.

3.21 Identification of product. Equipment, assemblies, and parts shall be marked for identification in accordance with MIL-STD-130.

3.22 Workmanship. The release, including all parts and accessories, shall be constructed and finished in a thoroughly workmanlike manner. Particular attention shall be given to freedom of parts from burrs and sharp edges.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of tests. The inspection and testing of releases shall be classified as follows:

a. Qualification testing . . . . . See 4.3

b. Quality conformance inspection . . . See 4.4

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T06:42Z
Check the source to verify that this is the current version before use.

4.3 Qualification testing.

4.3.1 Qualification test samples. The qualification test samples shall consist of two releases representative of the production equipment. The samples shall be identified with the manufacturer's part number and such other information as required by the procuring activity.

4.3.2 Qualification test report. When the tests are conducted at a location other than the laboratory of the qualifying activity, a test report conforming to MIL-STD-831 shall be furnished to the qualifying activity.

4.3.3 Qualification tests. The qualification tests shall consist of all tests described under 4.5.

4.4 Quality conformance inspection. Quality conformance inspection shall consist of:

a. Individual tests.

b. Sampling plans and tests.

* 4.4.1 Individual tests. Each release shall be subjected to the following tests as described under 4.5:

a. Examination of product.

b. Safety latch efficacy.

c. Timer accuracy.

d. Pressure sensing device accuracy.

(1) Type F-1B

(2) Type FF-1, Test Method A

e. Pull of spring.

4.4.2 Sampling plans and tests.

* 4.4.2.1 Sampling plan A. Releases shall be selected at random in accordance with MIL-STD-105 and subjected to the tests listed below and described under 4.5. For the Type FF-1, the inspection shall be General Inspection Level II and the acceptable quality level (AQL) 0.40 defects per hundred units. For the Type F-1B, the AQL shall be 1.5 percent defective. All releases subjected to these sampling tests shall be returned for rework and shall not be chosen for sampling tests for any future lot, but must repeat individual test requirements after rework.

a. High altitude - Low temperature.

b. High temperature.

c. Pressure sensing device hysteresis.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T06:42Z
Check the source to verify that this is the current version before use.

d. Arming pin pull.

e. Pressure sensing device accuracy for Type FF-1 (Test Method B).

4.4.2.2 Sampling plan B. Five releases shall be selected at random from every 500 or fraction thereof produced and subjected to the following tests as described under 4.5:

a. High temperature exposure (see 4.5.10).

b. High altitude - low temperature exposure (see 4.5.11).

c. Vibration.

d. Shock.

4.4.2.2.1 Rejection and retest. When an item selected from a production run fails to meet the specification, no items still on hand or later produced shall be accepted until the extent and cause of failure have been determined and appropriately corrected. The contractor shall explain to the Government representative the cause of failure and the action taken to preclude recurrence. After correction, all of the tests shall be repeated.

4.4.2.2.2 Individual tests may continue. For operational reasons, individual tests or other sampling plans may be continued pending the investigation of a sampling test failure. Final acceptance of the items on hand, or produced later, shall not be made until it is determined that all items meet all the requirements of the specification.

4.4.2.3 Defects in items already accepted. The investigation of a test failure could indicate that defects may exist in items already accepted. If so, the contractor shall fully advise the procuring activity of all the defects likely to be found and the method of correcting them.

4.5 Test methods.

4.5.1 Examination of product. The release shall be inspected to determine compliance with this specification with respect to materials, workmanship, dimensions and marking.

4.5.2 Safety latch efficacy. After the release has been operated in the normal manner, a check shall be made to determine the efficacy of the safety latches. It shall not be possible to extend the ripcord action cable or set the timing knob unless the main spring has been fully wound.

4.5.3 Timer accuracy. The release shall be checked at the 2-1/2 and 12-1/2-second settings to determine if the timer accuracy is within the limits specified in 3.8.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T06:42Z
Check the source to verify that this is the current version before use.

* 4.5.4 Pressure sensing device accuracy.

* 4.5.4.1 Type FF-1 test method A. The pressure sensing device shall be checked for accuracy at each of the altitude dial settings specified in Table II. Each altitude dial setting shall be checked by exposing the release in a test chamber at an altitude pressure level of a minimum of 2,000 feet above the altitude at which the timer is blocked. The release shall then be armed and the pressure in the test chamber increased until the timer activates. The timer activation points noted at each of the required altitude dial settings shall be within the limits specified in Table II. The altitude indicator, which serves as the pressure sensing device leak detector, shall begin to move at 3,000 feet and shall be within ±1,000 feet at 5,000 feet and within ±3,000 feet at 15,000 feet.

* 4.5.4.2 Type FF-1 test method B. With an altitude dial setting of 5,500 feet and the mechanism fully wound, the pressure sensing device shall be placed in a test chamber The internal chamber pressure shall be decreased from sea level (29.92 inches Hg) to a pressure altitude of 7,700 feet (22.48 inches Hg) and shall be stabilized at this point. After pressure stabilization at 7,700 feet, the release shall be armed by removing the arming pin. The release shall not be activated at this point.

* 4.5.4.3 Type F-1B pressure sensing device accuracy. The pressure sensing device shall be checked in an altitude test chamber by taking the test chamber above a present altitude of 14,000 feet and lowering it to determine the release point which shall be within the limits specified in 3.9. The altitude indicator, which serves as the pressure sensing device leak detector, shall begin to move at 3,000 feet and shall be within ±1,000 at 5,000 feet and within ±3,000 at 20,000 feet.

4.5.5 Pull of spring. The pull exerted by the spring of the release shall be determined after 2 inches of cable movement for the 2.5 second time setting and for the 12-1/2-second time setting. The pull exerted shall meet the requirements specified in 3 7.

4.5.6 High altitude - low temperature. The release shall be placed in a chamber and maintained at a temperature of approximately 54°C and a pressure equivalent to approximately 50,000 feet for a period of 2 hours. The release shall then be maintained at a temperature of approximately -54°C at atmospheric pressure for an additional period of 2 hours. Upon completion of the exposure and while at this temperature, the release shall be subjected to the following:

a. For Type FF-1 - Tests specified in 4.5.3 and 4 5.4.1 with the altitude dial set at 3,000 feet.

b. For Type F-1B - Tests specified in 4.5.3 and 4.5.4.3 with the altitude dial set at 5,000 feet.

The timer shall be set to 10 seconds for the tests. Upon completion of these tests the ripcord power cable assembly housing shall be bent 180 degrees over a 3-inch mandrel. The housing shall not crack or show visible damage.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T06:42Z
Check the source to verify that this is the current version before use.

4.5.7 High temperature. The release shall be subjected to a temperature of 71° ±2°C for a period of 2 hours. While at this temperature, the release shall be subjected to the following:

a. For Type FF-1 - Tests specified in 4.5.3 and 4.5.4.1 with the altitude dial set at 5,000 feet. The timer shall be set to 10 seconds.

b. For Type F-1B - Tests specified in 4.5.3 and 4.5.4.3. Unless otherwise specified (see 6.2), the altitude dial shall be set at 14,000 feet and the timer at 10 seconds.

* 4.5.8 Pressure sensing device hysteresis.

* 4.5.8.1 Type FF-1. The pressure sensing device hysteresis shall be checked by setting the altitude-setting knob on 3,000 feet and placing the release in an altitude chamber. Using the release point for the 3,000-foot setting, as determined in 4.5.4.1, the release shall be recocked and taken to an altitude of 30,000 feet and then lowered at a rate of 180 to 230 feet per second to again determine the release points. The altitude at which the timer is released shall be within the limits specified in 3.9b herein.

* 4.5.8.2 Type F-1B. The pressure sensing device hysteresis shall be checked by setting the altitude-setting knob on 14,000 feet and placing the release in an altitude chamber. Using the release point for the 14,000-foot setting, as determined in 4.5.4.3, the release shall be recocked and taken to an altitude of 30,000 feet and then lowered at a rate of 180 to 230 feet per second to again determine the release points. The altitude at which the timer is released shall be within the limits specified in 3.9a.

* 4.5.9 Arming pin pull. With a spring scale attached to the knob and the knob snapped in place in the ferrule, the force required to remove the knob from the ferrule shall be measured and shall be between 15 and 20 pounds. The arming pin knob shall continue to be pulled, and the force required to extract the arming pin from the release shall be measured and shall be between 10 and 20 pounds. Units supplied without arming cables shall be subjected to the arming pin pull utilizing a hardened steel test pin made in accordance with dimensions shown on Drawing AF-1218. Under a straight pull, a measured force between 10 and 20 pounds shall be required to extract the arming pin from the release.

4.5.10 High temperature exposure. The test specified in 4.5.7 shall be repeated, except that the time of exposure shall be 15 hours.

4.5.11 High altitude - low temperature exposure. The test specified in 4.5.6 shall be repeated except that the time of exposure for the -54°C, 50,000 feet period shall be 48 hours. For the -54°C, ambient atmospheric pressure period, the time of exposure shall remain at 2 hours.

* 4.5.12 Vibration. The release, cocked and with the arming pin inserted, shall be subjected to linear vertical vibration for a period of 30 minutes in each of three mutually perpendicular planes on a suitable vibration stand. When mounted in the horizontal plane, the release shall always be mounted upside down. In one of the positions, the release shall be so mounted that the pawl for the reel release shall have the additional force of gravity tending to trip out this pawl. During this period of vibration, the frequency shall be varied continuously from 10 to 55 cycles per second with a double amplitude

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T06:42Z
Check the source to verify that this is the current version before use.

(total excursion) of 0.030 inch. The release shall not operate the ripcord power cable during the period of vibration.

a. For Type FF-1, the pressure sensing device shall be set for 5,000 feet, the release shall then be placed in an altitude chamber and subjected to a pressure corresponding to 10,000 feet. The release shall be vibrated for at least 30 minutes in each plane. Upon completion of the vibration in each plane, the arming pin shall be pulled and the altitude brought down to ascertain the altitude at which the release operates. The tests shall be repeated with the chamber at 20,000 feet and with the aneroid set for 15,000 feet.

b. For Type F-1B, the pressure sensing device shall be set for 10,000 feet, the release shall then be placed in an altitude chamber and subjected to a pressure corresponding to 15,000 feet. The release shall be vibrated for at least 30 minutes in each plane. Upon completion of the vibration in each plane, the arming pin shall be pulled and the altitude brought down to ascertain the altitude at which the release operates. The tests shall be repeated with the chamber at 25,000 feet and with the pressure sensing device set for 20,000 feet.

After all vibration tests, the release shall undergo the tests specified in 4.5.2 through 4.5.5, except 4.5.4.2, noting particularly whether the timer runs down without operating the ripcord. Any looseness in the mechanism or damage to any part of the release resulting from the vibration shall be cause for rejection.

* 4.5.13 Shock. The release shall be mounted on sufficient mass and dropped through such a height that when decelerated by resilient impact a deceleration of 30-g shall be obtained. The release, cocked and with the arming pin inserted, shall be tested with the axis mounted in three different planes. The release shall not pull the power cable during these tests. With the aneroid set for 14,000 feet for the Type F-1B or 3,000 feet for the Type FF-1, the release shall be mounted with the cover down in a horizontal plane on a shock-testing machine in an altitude chamber. At an altitude of 4,000 feet above the elevation at which the release theoretically trips, the arming pin shall be removed. While still at this altitude, the release shall be subjected to a 30-g shock which shall not trip the power cable. The test shall be repeated for each of two additional positions at right angles to this plane and three positions opposite the foregoing, and the release shall not be tripped by a 30-g shock applied under the same conditions. Any calibrated shock testing machine capable of producing the specified shock when decelerated by resilient impact is considered acceptable for conduct of the test (see 6.5). Following this test the release shall undergo the tests specified in 4.5.2 through 4.5.5 except 4.5.4.1. Any looseness in the mechanism or damage to any part of the release resulting from this test shall be cause for rejection.

4.5.14 Acceleration. The release shall be mounted on the apparatus (centrifuge), and the apparatus shall be operated at a speed that will produce an acceleration of 30 g. The release shall be mounted first in a position parallel to the escapement assembly (usually horizontal) on the center of the turntable which is secured to the mounting platform of the centrifuge. The applied acceleration of 30 g shall be attained, stabilized, and maintained for a period of not less than 1 minute for each successive position. At the end of the 1-minute period, the arming pin shall be pulled. The release shall operate under these conditions, and the apparatus (centrifuge) shall be slowed. The release shall then be checked and reset for the next test. The release shall

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T06:42Z
Check the source to verify that this is the current version before use.

be subjected to tests where the release is mounted in positions that will allow a 30-g force to be applied in the plus or minus direction along the X, Y, and Z axis and the release set in six different positions for this axis. If any of these tests prove doubtful the specific test shall be re-run. Failure of the release to operate shall be cause for rejection.

4.5.15 Sand and dust. Two releases shall be subjected to sand and dust in accordance with procedure I of MIL-E-5272. The releases shall then be subjected to the tests specified in 4.5.2 through 4.5.5, except 4.5.4.2. There shall be no evidence of sand and dust within the releases.

* 4.5.16 Humidity. Two releases shall be subjected to humidity in accordance with procedure I of MIL-E-5272 except that the number of cycles shall be limited to five. At the completion of the test, the releases shall be inspected for collection of moisture in the interior of the case, corrosion of metal parts, or other damage. At the completion of the fifth cycle, the releases shall be subjected to the tests specified in 4.5.2 through 4.5.5, except 4.5.4.2. The timer and pressure sensing device errors shall not exceed the provisions of this specification relating to such errors.

4.5.17 Salt spray. Two releases shall be subjected to salt spray in accordance with MIL-E-5272 for a period of 50 hours. At the end of this period, the releases shall be subjected to the tests specified in 4.5.2 through 4.5.5, except 4.5.4.2.

4.5.18 Venting test. With the ripcord power cable removed from the cable guides and with the arming pin removed, a rubber tube shall be placed over the ripcord cable guide or connected to a pressure part which will reflect the internal case pressure; the other end of the tube shall be connected to an altimeter. Both the release and the altimeter shall be placed in an altitude chamber. The pressure shall be varied to simulate 25,000 feet altitude, then increased to a rate equivalent to not more than 250 feet per second nor less than 150 feet per second. The difference in the reading of this altimeter and the chamber altitude as the pressure is reduced shall not exceed ±100 feet. Two samples shall be subjected to this test.

* 4.5.19 Pressure sensing device cycling. Prior to this test, two releases shall be subjected to the test specified in 4.5.4.1 or 4.5.4.3, as applicable. With the arming pins inserted, the releases shall then be subjected to continuously varying altitude cycles at room temperature, from sea level to 35,000 feet and back to sea level. Upon completion of 1,000 cycles, a pressure sensing device accuracy check shall again be made. The accuracy shall be within the requirements specified in 3.9.

4.5.20 Overpressure. Two releases shall be subjected to an absolute pressure of 50 inches mercury for a period of 1 hour. For a period of not less than 6 hours following this overpressure, the releases shall not be subjected to any operation other than atmospheric pressure. An accuracy test shall then be made on the releases, and the accuracy shall meet the requirements specified in 3.9.

4.5.21 Cable housing and ferrules. With the end ferrules of the cable gripped in a fixture that will apply load uniformly to the cable housing, the housing shall be subjected to a proof load in tension of 150 pounds.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T06:42Z
Check the source to verify that this is the current version before use.

4.5.22 Strength of knob material. Two releases shall be tested with a 100-pound weight which shall be suspended from the arming pins. With the knob used as the point of support, the ball on the end of the arming-pin cable shall not shear or tear the material composing the seat for this ball.

4.5.23 Life test. The power cable, which shall be protected by the power cable housing, is to be attached to a spring by a means that is capable of maintaining the load for the entire firing cycle. The release shall then be operated at room conditions a total of 70 times. After the 70 operations, the release shall be subjected to the tests specified in 4.5.2 through 4.5.5, except 4.5.4.2.

4.5.24 Crush load. The cover assembly shall be subjected to a crush load to determine compliance with 3.12. A 3/8-inch diameter steel rod having a flat face shall be used to produce the crush load.

4.6 Inspection of preparation for delivery requirements. Sample items or packs and the inspection of the preservation, packaging, packing, and marking for shipment and storage shall be in accordance with the requirements of Section 5, or the documents specified therein.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Preservation and packaging shall be level A or C as specified (see 6.2).

5.1.1 Level A. Unless otherwise specified in the contract or order, each release assembly shall be preserved and packaged in accordance with MIL P-116, method ICI (without preservative compound), one each overboxed in a unit container conforming to PPP-B-676.

5.1.2 Level C. Releases shall be preserved and packaged in a manner that will afford adequate protection against corrosion, deterioration, and physical damage during shipment from supply source to the first receiving activity for immediate use. This level may conform to the supplier's commercial practice provided the latter meets the requirements of this level.

5.2 Packing. Packing shall be level A, B or C as specified (see 6.2).

5.2.1 Level A. Release assemblies preserved and packaged as specified in 5.1.1, shall be packed in a snug-fitting fiberboard shipping container conforming to style RSC, grade V2s of PPP-B-636; or overseas type, style A or J, type 2 load of PPP-B-601, as specified (see 6.2). Each shipping container shall be closed and reinforced in accordance with the appendix of the applicable container specification.

5.2.2 Level B. Release assemblies preserved and packaged as specified in 5.1.1, shall be packed in a snug-fitting fiberboard shipping container conforming to style RSC, type CF (variety SW) or SF, class domestic, grade 275 of PPP-B-636. Each shipping container shall be closed in accordance with method II as specified in the appendix

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T06:42Z
Check the source to verify that this is the current version before use.

5.2.2.1 When specified (see 6.2), the shipping container shall be a grade V3c or V3s fiberboard box fabricated in accordance with PPP-B-636 and closed in accordance with the appendix of the container specification.

5.2.3 Level C. Level C packing shall afford adequate protection at the lowest rate against damage during direct domestic shipment from the supply source to the receiving activity for immediate use. This level shall conform to applicable carrier rules and regulations and may be the supplier's commercial practice, provided the latter meets the requirements of this level.

5.3 Physical protection. Cushioning, blocking, and bracing shall be in accordance with MIL-STD-1186 except that waterproofing requirements for cushioning materials and containers for domestic shipments shall be waived. Drop tests of MIL-STD-1186 shall be waived when preservation, packaging and packing of the item are for immediate use or when the drop tests of MIL-P-116 are applicable.

5.4 Marking for shipment and storage. Interior and exterior containers shall be marked in accordance with MIL-STD-129. The nomenclature shall be as follows: Release Parachute Ripcord, Automatic, Type F-1B or FF-1, Aneroid Timer Control, With Altitude Indicator.

5.4.1 Reinspection date markings. Reinspection marking shall be as specified by the procuring activity.

6. NOTES

6.1 Intended use. The types F-1B and FF-1, automatic parachute-ripcord releases covered by this specification are intended for use to automatically pull the ripcord of a parachute.

6.2 Ordering data. Procurement documents should specify the following:

a. Number, date, and title of this specification.

b. Type and class required (see 1.1 and 1.2).

c. Arming cable lengths (see 3.10.1).

d. Whether anchor is required (see 3.13).

e. Whether snap hook lanyard is required (see 3.14).

f. Whether lanyard is required (see 3.15).

g. Altitude dial and timer settings for type F-1B High Temperature Test, if other than specified (see 4.5.7b).

h. Applicable levels of preservation, packaging, and packing (see 5.1 and 5.2).

i. Type of shipping container desired for level A packing (see 5.2.1).

j. When weather-resistant grade fiberboard shipping containers are required for level B packing (see 5.2.2.1).

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T06:42Z
Check the source to verify that this is the current version before use.

6.3 Provisions for qualification. With respect to products requiring qualification, awards will be made only for products which are at the time set for opening of bids, qualified for inclusion in the applicable Qualified Products List whether or not such products have actually been so listed by that date. The attention of the suppliers is called to this requirement, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. The activity responsible for the Qualified Products List is the San Antonio Air Materiel Area, Service Engineering Division, ATTN: SANE, Kelly AFB, Texas 78241, and information pertaining to qualification of products may be obtained from that activity.

* 6.4 Arming knob assembly. Arming knob assemblies manufactured by U.S. Gauge Division of Ametek, Inc. as their part number of AF1219 have been found to meet the requirements of 3.10.2.

* 6.5 The mechanism in accordance with method 202B of MIL-STD-202C is suitable for conducting these tests except that the calibrated spring shall have a spring constant (K) of 5,000 to 5,500 pounds per inch. Copies of MIL-STD-202, Revision C are available from the Naval Publications and Forms Center, 5801 Tabor Ave., Philadelphia PA 19120.

* 6.6 The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodian:
Air Force - 82
Army - GL

Review Acitvities:
Air Force - 11
Army - GL, AV

Preparing Activity:
Air Force - 82

Project No. 1670-0490

☆ U. S. GOVERNMENT PRINTING OFFICE: 1974-603-121/355

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T06:42Z
Check the source to verify that this is the current version before use.

SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T06:42Z
Check the source to verify that this is the current version before use.

FOLD

POSTAGE AND FEES PAID

DEPARTMENT OF THE AIR FORCE

DOD 318

OFFICIAL BUSINESS

SAN ANTONIO AIR MATERIEL AREA
SERVICE ENGINEERING DIVISION (MTEOP)
KELLY AFB TX 78241

FOLD

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T06:42Z
Check the source to verify that this is the current version before use.